

Decoupling Stem-Cell Maintenance from Cellular Senescence

A Comparative Investigation of Meristem Longevity and Quality Control in *Ginkgo biloba* | Executive Handout

1. Executive Summary & Project Rationale

Human adult somatic stem cells undergo progressive exhaustion, mutational decay, and senescent cell-cycle arrest over decades. In contrast, centenarian gymnosperms such as *Ginkgo biloba* maintain active shoot apical meristems (SAM) and vascular cambium stem cells for over 1,000 years without age-related functional decay. This research project establishes a comprehensive 12-Step Research Framework to map the molecular circuits driving perennial stem-cell longevity and evaluate their cross-kingdom translatability to human cellular aging models.

2. Summary of the 3 Core Research Pillars

Pillar	Biological Scope & Steps	Key Molecular Mechanisms
Pillar I: Meristem Architecture	Steps 1–4: SAM & Cambium Spatial Signaling	WUS-CLV spatial feedback loop, Cytokinin/Auxin 40-fold WUS induction via AHK2/4, Apical hypoxic gradient, G-proteins (CT2, SIGB1).
Pillar II: Quality Control	Steps 5–8: Genomic Integrity & Proteostasis	Ku70/80 NHEJ ring scaffold (100% basic channel conservation), Polycomb H3K27me3 silencing, ATG8/NBR1 autophagic flux, HSP chaperones.
Pillar III: Translation	Steps 9–12: Age Gradients & Rescue	Tree-NEB single-nuclei protocol across 15y, 200y, and 600+y cohorts; pLenti-GbKu70 CRISPR rescue of human XRCC6-KO cell lines.

3. Experimental Blueprint: Hypothesis 4 (GbKu70 Human Rescue)

Objective: Test whether gymnosperm *GbKu70* can physically assembly with human Ku80 (XRCC5) and rescue double-strand break (DSB) repair kinetics in human *XRCC6*-KO cell lines.

- **Lentiviral Vector:** pLenti-EF1a-GbKu70-3xFLAG-P2A-Puro carrying human-codon-optimized *GbKu70* with wobble-base mutations.
- **CRISPR KO & Co-IP:** Electroporate Cas9-RNPs targeting human *XRCC6* in HEK293T/HeLa cells. Perform anti-FLAG Co-IP to confirm physical binding to human Ku80.
- **Repair Kinetics Readouts:** Quantify γ -H2AX / 53BP1 foci clearance half-life ($t_{1/2}$) and clonogenic survival (0–8 Gy) post-irradiation.

4. Itemized Budget & 3-Year Milestone Schedule

Category	Budget (\$)	3-Year Major Deliverables
Personnel	\$450,000	1 Postdoctoral Fellow (3Y), 1 Bioinformatics Associate (3Y), 0.20 FTE PI Effort.

Sequencing & Tree-NEB	\$280,000	10x Chromium snRNA-seq across 15y, 200y, and 600+y Ginkgo age cohorts.
Reagents & Cell Culture	\$175,000	Lentiviral packaging, CRISPR Cas9 RNPs, Bafilomycin A1, γ -H2AX antibodies.
Indirects & Overhead	\$100,000	Facility operations, institutional compliance, and administrative overhead.